

Revision of Linear Algebra and Probability

Advanced Statistical Inference

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Linear Algebra

1. Compute the determinant of the following matrix:

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

2. Find the eigenvalues and eigenvectors of the matrix:

$$B = \begin{pmatrix} 4 & 1 \\ 2 & 3 \end{pmatrix}$$

3. Prove that the set of vectors $\{(1, 2), (2, 4)\}$ is linearly dependent. (Hint: think about the determinant of the matrix formed by these vectors.)
4. Given the matrix $C = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, compute its inverse, if it exists.
5. Show that the trace of a square matrix is equal to the sum of its eigenvalues. (Hint: use the spectral decomposition and a property of the trace function.)
6. Given the following system of equations, represent it in matrix form:

$$4x + 2y + 2z = 4$$

$$2x + 2y + 1z = 2$$

$$2x + 1y + 2z = 2$$

7. Solve the system of equations from the previous question using Cholesky decomposition and the forward/backward substitution method.

- What is the connection between the diagonal elements of the Cholesky decomposition of a positive definite matrix and the eigenvalues of that matrix? (Hint: write the determinant of the matrix in terms of its Cholesky decomposition and in terms of its eigendecomposition.)

Probability

- Let X be a continuous random variable with probability density function (pdf):

$$f(x) = \begin{cases} 2x & \text{if } 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Is $f(x)$ a valid pdf? Justify your answer.

- Show that the sum of expectations is equal to the expectation of the sum, i.e., for random variables X and Y , prove that:

$$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$$

- Given the probability density function of a normal distribution:

$$p(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

write down the expression for $\log p(x)$ and simplify it as much as possible.

- Given the probability density function of multivariate normal distribution:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{d/2} \det(\mathbf{\Sigma})^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \boldsymbol{\mu})^\top \mathbf{\Sigma}^{-1}(\mathbf{x} - \boldsymbol{\mu})\right)$$

write down the expression for $\log p(\mathbf{x})$ and simplify it as much as possible.

- From the previous question, write down a numerically stable expression for $\log p(\mathbf{x})$ by avoiding the computation of the determinant and the inverse of the covariance matrix $\mathbf{\Sigma}$ directly. (Hint: start from the Cholesky decomposition of $\mathbf{\Sigma}$)
- Given 3 random variables X , Y , and Z , write down all possible expressions to compute the joint probability $p(X, Y, Z)$ using the chain rule of probability.
- Assume the $\mathbf{x}$ is distributed according to a multivariate normal distribution with mean $\boldsymbol{\mu} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$ and covariance matrix $\mathbf{\Sigma} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 2 \end{pmatrix}$. Let $\mathbf{A} = \begin{pmatrix} 1 & 0.5 \\ 0.5 & 1 \end{pmatrix}$ be a linear transformation matrix. Find the distribution of the transformed variable $\mathbf{y} = \mathbf{A}\mathbf{x}$.

8. Let $\mathbf{x}$ be a random vector distributed according to a multivariate normal distribution with mean $\boldsymbol{\mu} = \begin{pmatrix} 1 & 2 & 3 \end{pmatrix}^\top$ and covariance matrix

$$\boldsymbol{\Sigma} = \begin{pmatrix} 6 & 1 & 2 \\ 1 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

- Compute the marginal distribution $p(x_1, x_3)$.
- Compute the conditional distribution $p(x_1 \mid x_2 = 2, x_3 = 4)$.
- What can you say about the marginal distribution $p(x_2, x_3)$? Are x_2 and x_3 correlated?